

Teen Mental Health & Social Media Usage Analysis Report

Teen Mental Health Analysis: Understanding the Impact of Social Media Usage on Psychological Well-being.

1. Executive Summary

This study analyses the relationship between social media usage and mental health among teenagers using a dataset of **1,200 respondents** aged **13–19 years**.

The analysis investigates how factors such as:

- Daily social media usage
- Sleep duration
- Screen time before sleep
- Physical activity
- Stress levels
- Anxiety levels
- Social interaction

influence mental health outcomes, particularly depression.

Key Findings

- Teens spend an average of **4.54 hours daily** on social media.
- Depression prevalence in the dataset is **2.58%**.
- Depressed teens sleep significantly less (**4.74 hours**) compared to non-depressed teens (**6.48 hours**).
- Depressed teens show much higher stress (**8.48/10**) and anxiety (**8.61/10**) levels.
- Higher daily social media usage is associated with increased depression risk.
- Sleep duration exhibits the strongest negative relationship with depression.

2. Dataset Overview

Dataset Size

Metric	Value
Total Records	1,200
Features	13
Missing Values	0

Metric	Value
Age Range	13–19 years

Variables

Variable	Description
AGE	Age of participant
GENDER	Male/Female
DAILY SOCIAL MEDIA HOURS	Average hours spent daily
PLATFORM USAGE	Instagram, TikTok, Both
SLEEP HOURS	Average daily sleep
SCREEN TIME BEFORE SLEEP	Hours before bedtime
ACADEMIC PERFORMANCE	Academic rating
PHYSICAL ACTIVITY	Physical activity level
SOCIAL INTERACTION LEVEL	Low, Medium, High
STRESS LEVEL	1–10 scale
ANXIETY LEVEL	1–10 scale
ADDICTION LEVEL	1–10 scale
DEPRESSION LABEL	0 = No Depression, 1 = Depression

3. Data Quality Assessment

Missing Values

Category	Count
Missing Values	0

The dataset is clean and ready for analysis.

Duplicate Records

No significant duplicate issues observed.

Data Types

- Numerical Variables: 10
- Categorical Variables: 3

4. Demographic Analysis

Age Distribution

Statistic	Value
Minimum Age	13
Maximum Age	19
Average Age	15.93

The sample is fairly distributed across teenage years.

Gender Distribution

Gender Count Percentage

Male	615	51.3%
Female	585	48.7%

The dataset is balanced by gender.

5. Social Media Usage Analysis

Platform Preference

Platform	Users
Instagram	411
TikTok	398
Both Platforms	391

Instagram is the most frequently used platform.

Daily Social Media Hours

Summary Statistics

Metric	Value
Mean	4.54 Hours
Median	4.50 Hours
Minimum	1 Hour
Maximum	8 Hours

Interpretation

Teenagers spend approximately half of their waking leisure time on social media platforms.

6. Sleep Behaviour Analysis

Sleep Hours

Statistic	Value
Average Sleep	6.43 Hours
Median Sleep	6 Hours
Maximum	9 Hours

Insight

The average sleep duration is below the recommended 8–10 hours for teenagers.

7. Mental Health Analysis

Stress Level

Statistic	Value
Average Stress	5.45
Median Stress	5
Maximum	10

Anxiety Level

Statistic	Value
Average Anxiety	5.64
Median Anxiety	6
Maximum	10

Addiction Level

Statistic	Value
Average Addiction	5.56
Median Addiction	6
Maximum	10

8. Depression Analysis

Overall Depression Rate

Status	Count	Percentage
No Depression	1,169	97.42%
Depression	31	2.58%

The majority of participants do not exhibit depression symptoms.

9. Depression vs Key Variables

Social Media Usage

Group	Avg Hours
Non-Depressed	4.48
Depressed	6.72

Finding

Depressed teenagers spend approximately:

50% more time on social media
than non-depressed teenagers.

Sleep Hours

Group	Avg Sleep
Non-Depressed	6.48
Depressed	4.74

Finding

Depressed teens sleep nearly:

1.74 hours less per day
than non-depressed teens.

Stress Levels

Group	Avg Stress
Non-Depressed	5.37
Depressed	8.48

Finding

Stress is dramatically higher among depressed individuals.

Anxiety Levels

Group	Avg Anxiety
Non-Depressed	5.56
Depressed	8.61

Finding

Anxiety appears strongly linked with depression.

Physical Activity

Group	Avg Activity
Non-Depressed	1.03
Depressed	0.90

Finding

Lower physical activity is associated with poorer mental health.

10. Gender-Based Analysis

Depression Rate by Gender

Gender	Depression Rate
Female	2.91%
Male	2.28%

Finding

Female participants show slightly higher depression prevalence.

11. Platform-Based Analysis

Depression Rate by Platform

Platform	Depression Rate
TikTok	3.02%
Instagram	2.43%
Both	2.30%

Finding

TikTok users demonstrate the highest depression rate in this dataset.

12. Social Interaction Analysis

Interaction Level	Depression Rate
High	2.71%
Medium	2.88%
Low	2.17%

The relationship between social interaction and depression appears weak in this dataset.

13. Insights

Key Risk Indicators

The strongest warning signs of depression include:

- Excessive social media use
- Reduced sleep duration
- High stress levels
- High anxiety levels
- Low physical activity

Recommended Interventions

For Schools

- Mental health awareness programs
- Sleep hygiene education
- Digital wellness campaigns

For Parents

- Monitor screen time
- Encourage outdoor activities
- Promote healthy sleep routines

For Policymakers

- Social media literacy programs
- Mental health support initiatives
- School counselling enhancements

14. Conclusion

This analysis reveals a clear relationship between **social media behaviour and mental health outcomes among teenagers**.

The most important findings are:

1. Higher social media usage is associated with greater depression risk.
2. Sleep deprivation is the strongest negative factor affecting mental health.
3. Stress and anxiety levels rise substantially among depressed teens.
4. Physical activity appears to offer protective benefits.
5. Female and TikTok users show slightly higher depression prevalence.

Overall, the study suggests that **balanced social media use, adequate sleep, and active lifestyles are critical factors in maintaining adolescent mental well-being**.